

AIRLINE MANAGEMENT SYSTEM

Full Research Explanation for Presentation

1. Research Paper: Airline Management Information System at Arkia Israeli Airlines

Authors: Israel Borovits and Seev Neumann

Year: 1988

Journal: MIS Quarterly

Volume: 12, Issue 1

Pages: 127–137

DOI: 10.2307/248813

This is the most directly relevant research paper for our project. It presents an approach for developing and implementing a Management Information System for a small airline. The research describes computerized, real-time support for airline sales and reservation activities and discusses the role of information systems in airline management.

Main idea: Airline operations generate large amounts of information. A computerized and integrated information system can help an airline organize and use that information more effectively.

Connection with our project: Our Airline Management System applies the same broad information-management concept to a B.Tech-level internal system. Our staff-oriented system can manage flight information, passenger records, aircraft capacity, scheduling and operational status.

Presentation line:

“The most relevant research paper for our project is *Airline Management Information System at Arkia Israeli Airlines* by Borovits and Neumann. It discusses the development and implementation of a computerized management information system for an airline. Based on this concept, our project focuses on centralizing important airline information for management staff.”

2. Research Paper: Airline Reservations Systems — Lessons from History

Authors: Duncan G. Copeland and James L. McKenney

Year: 1988

Journal: MIS Quarterly

This research examines the development and role of airline reservation systems. Reservation systems have to manage large amounts of airline information and support activities such as reservations and related data handling.

Main idea: Computerized reservation systems make it possible to manage airline information more systematically than purely manual processes.

Connection with our project: This research supports the passenger and booking-record side of our system. We can maintain passenger-related information and connect it with the relevant flight records.

Presentation line:

“We also studied research on airline reservation systems. This helped us understand the importance of computerized management of passenger and reservation-related information, which supports our passenger and booking-record modules.”

3. Research Paper: Airline Timetable Development and Fleet Assignment

Authors: Keji Wei, Vikrant Vaze and Alexandre Jacquillat

Published online: December 23, 2019

Journal: Transportation Science

Volume: 54, Issue 1

Pages: 139–163

DOI: 10.1287/trsc.2019.0924

This research focuses on airline timetable development and aircraft fleet assignment. It explains that flight scheduling and aircraft assignment are strongly interconnected decisions. The study uses an advanced optimization formulation and computational case-study analysis.

Main idea: An airline must decide when flights should operate and which aircraft should operate those flights. These decisions influence airline operations and passenger choices.

Connection with our project: We use the basic management concept in our Flight Scheduling and Aircraft Management modules. Our project does not attempt to reproduce the advanced mathematical optimization used in the research.

Presentation line:

“The third research paper focuses on airline timetable development and fleet assignment. It helped us understand that flight scheduling and aircraft allocation are interconnected parts of airline operations. We use this concept in a simplified form in our project.”

4. What We Learned From the Three Papers

- Paper 1 shows how computerized information systems can support airline management.
- Paper 2 shows the importance of computerized reservation and passenger information management.
- Paper 3 shows the relationship between flight scheduling and aircraft assignment.
- Together, the papers show that airline operations depend on organized and interconnected information.
- Our project combines the core concepts into a simplified internal management system.

5. Research Framework

Research Papers → Airline Information Systems + Reservation Systems + Scheduling/Fleet Assignment → **Core Airline Management Requirements** → **Our Internal Airline Management System**

6. Research Gap

A research gap does not mean that nobody has ever developed an airline management system. Existing airline systems and research already cover management information, reservations, scheduling, optimization and many other functions. The gap for our academic project is the practical translation of these concepts into a simplified, centralized and staff-oriented system that can realistically be implemented and demonstrated as a B.Tech project.

Professional explanation:

“Existing commercial airline systems are large and complex. Our project focuses on a simplified internal management system containing the core functions required by airline management staff, making it practical and suitable for academic implementation.”

7. Research Methodology

1. Identify the airline-management problem.
2. Search and study published research.
3. Select papers related to management information systems, reservations and scheduling.
4. Identify important concepts from the literature.
5. Map those concepts to our project requirements.
6. Define the core system modules.
7. Implement a simplified practical solution.

8. Why Did We Select These Papers?

“We selected these papers because together they cover the three major research areas related to our project. The Borovits and Neumann paper provides the airline management information-system foundation. The Copeland and McKenney paper provides background on airline reservation systems. The Wei, Vaze and Jacquillat paper provides research related to flight timetabling and aircraft fleet assignment.”

9. Contribution of Our Project

Our project is not claiming to invent a new airline algorithm. Its contribution is the practical implementation of core airline information-management concepts in a simplified internal system designed for airline management staff.

Our system brings together:

Flight Management + Passenger Records + Aircraft/Capacity + Scheduling + Operational Information

10. What Is Novel About Our Project?

“The focus of our project is an internal staff-oriented management system rather than a customer-facing airline booking platform. We concentrate on integrating core operational records into a simple centralized application suitable for academic implementation.”

11. Two-Minute Full Research Explanation

“For our Airline Management System, we conducted a literature study of existing research related to airline information systems, airline reservation systems and airline scheduling. The first and most relevant paper we studied was *Airline Management Information System at Arkia Israeli Airlines* by Israel Borovits and Seev Neumann, published in MIS Quarterly in 1988. The paper discusses the development and implementation of a computerized management information system for an airline. This research helped us understand how information systems can support

airline management and operations. The second research area we studied was airline reservation systems. This helped us understand the importance of managing passenger and reservation information electronically. The third research paper focuses on airline timetable development and fleet assignment. It helped us understand the relationship between flight schedules and aircraft allocation. Based on these studies, we identified the core requirements for our project. Our system focuses on flight management, passenger records, aircraft capacity, scheduling and operational information. The research gap is not that airline management systems do not exist. Instead, commercial systems are much more complex than what is practical for an academic project. Therefore, our project provides a simplified internal management system focused on the core needs of airline management staff. The main contribution is the practical implementation of these airline information-management concepts in one centralized system.”

12. One-Sentence Research Conclusion

“Our research studied how information systems, reservation systems and scheduling technologies support airline operations, and we applied their core concepts to develop a simplified internal Airline Management System.”

13. References

1. Borovits, I., & Neumann, S. (1988). *Airline Management Information System at Arkia Israeli Airlines*. MIS Quarterly, 12(1), 127–137. DOI: 10.2307/248813.
2. Copeland, D. G., & McKenney, J. L. (1988). *Airline Reservations Systems: Lessons from History*. MIS Quarterly.
3. Wei, K., Vaze, V., & Jacquillat, A. (2019/2020). *Airline Timetable Development and Fleet Assignment Incorporating Passenger Choice*. Transportation Science, 54(1), 139–163. DOI: 10.1287/trsc.2019.0924.